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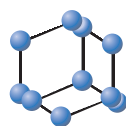
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Herbal Drugs and Natural Products in the light of Nanotechnology and Nanomedicine for Developing Drug Formulations

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Abstract: Natural products and medicinal plants have played a vital role in providing healthcare and ensuring well-being for many civilizations since antiquity. It is estimated that around 50% of drugs in the market have a natural product origin especially medicinal plants and herbal drugs, animals, fungi, and marine organisms. Some of these biologically active constituents of extracts have low absorption and distribution which, as a result, lead to loss of bioavailability and efficacy and might hamper their applications in the clinic. To overcome these impediments for the formulation of herbal drugs, food supplements, and essential oils, several nanomedical approaches such as liposomes, microemulsions, polymeric nanoparticles, solid lipid nanoparticles (SLNs), liquid crystal systems (LC), and precursor systems for liquid crystals (PSLCs) have been proposed. Nanoparticles have been used to modify and ameliorate the pharmacokinetic and pharmacodynamic properties of different drugs, thus incorporating biotechnological systems may be useful to enhance the bioavailability and bioactivity of herbal drug formulations. Consequently, essential for any natural compounds is the extent of its absorption after being ingested and its ability to be distributed in various tissues or organs of the body. The present review article aims to give an overview of the recent advancements in developing herbal drug formulations based on nanoparticle technologies.

Keywords: Bioavailability, bioactivity, nanomedicine, phytomedicine, drug delivery, natural products.

1. INTRODUCTION

There is an increasing demand for natural compounds, especially in the pharmaceutical, food, and cosmetic industries, to provide healthcare and ensure well-being for human beings. It has been estimated that nearly 200,000 natural compounds derived from higher plants, animals, fungi, and marine organisms are currently known, and have been investigated in traditional and complementary medicines as well as in modern drug discovery [1]. Currently, despite the availability of chemical drugs from the pharmaceutical industry to supply patients with medicines, a huge segment of the population continues to take advantage of integrative medicine in many developed and developing countries [2-4]. The World Health Organization Traditional Medicine (TM) Strategy 2014-2023 proclaimed that for many millions of people folk treatments, traditional practitioners, as well as herbal medicines, are the predominant source of the healthcare system, if not the only source [5].

The prevalent sources of natural compounds are medicinal and aromatic plants that have been widely used for curing or preventing a particular symptom of disease for many centuries in different traditional medicinal systems all over the world [4, 6]. Moreover, they are often used in the production of herbal drug formulations as they are considered to be safe as compared to modern conventional medicine and have economic value *via* international trade [4]. It is believed that only 5-15% of approximately 250,000 plant species have been investigated for medicinal purposes globally [1], and only 6% have been screened for their biological components [7]. About 50 % of prescribed and Over The Counter (OTC) medications are derived directly or indirectly from products of natural origins [8]. Bioactive compounds are a large variety of chemical substances, which are mainly secondary metabolites that are synthesized by an organism; many of these metabolites are found to be unique to a particular organism, which plays no role in basic life functions such as growth or reproduction [9].

In the past decades, many research studies on phytochemicals focused on plant-derived lead compounds [7].

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Nowadays, elucidating the chemical composition, vast structural diversity, their physicochemical, biological, and pharmacological properties, the potential mechanism of action, safety evaluation and toxicological studies of natural products has become a popular research focus for all researchers and scientific communities, and this may lead to the innovation of natural products with fewer adverse effects than present drugs [2, 3]. For instance, Pharmacognosy has succeeded in the isolation of paclitaxel or taxol from the bark of the Pacific yew tree *Taxus brevifolia*. Nutt is now a drug approved by the Food and Drug Administration (FDA) for ovarian and breast cancers [10], and artemisinin from *Artemisia annua* L is recommended by the World Health Organization (WHO) for treating uncomplicated malaria [11]. In this paper, we summarize the up-to-date knowledge pertaining to herbal drug formulations and food supplements based on nanoparticle technologies (Fig. 1).

2. BOTANICAL DRUG FORMULATIONS AND THEIR CHALLENGES

Plants have been empirically exploited and consumed by humans as medicines and foods due to their potential to prevent and ameliorate illnesses as well as in maintaining and promoting health [12]. Phytopharmacological science has already started to investigate the bioactivity of secondary metabolites (secondary products or natural products) present in medicinal plants such as phenols, terpenoids, nitrogen-containing alkaloids, and sulfur-containing compounds. These molecules, which can be fat-soluble (nonpolar) and water-soluble (polar), are organic compounds derived from primary metabolism in plants and are used as medicines, flavorings, and recreational drugs [13]. The production of these natural products is often produced in low concentrations. Their quality and quantity, as well as bioactivity, may vary according to several factors including botanical species, chemotypes, plant parts, storage, temperature, humidity, soil, and time of harvest. The major challenge (Fig. 2) for the delivery of several herbal drug preparations is poor solubility, poor oral bioavailability, low permeation, and poor stability [12], which might hamper their applications in the clinic [14]. Phytomedicines offer impressive activity in-vitro but less efficacy in-vivo, owing to their poor water solubility, lipophilicity and inappropriate molecular size resulting in poor absorption and hence poor physiologic availability [15]. Therefore, drug delivery vehicles, have to simultaneously ameliorate the solubility of the drug (herbal), diminish the degradation process, decrease any toxic effect, and mask any bad taste, while controlling the biological availability of the active substance and its response [16]. Unfortunately, some of them like flavonoids, tannins, and terpenoids are not absorbed in adequate concentration owing to poor membrane permeability, rapid clearance and enzymatic degradation, causing loss of bioavailability and efficacy [13, 3].

The effectiveness of botanical and herbal species generally relies on the availability of active compounds, and this should be delivered by carriers during the whole treatment period directly toward the desired target organs in sufficient concentration, which are not completely obtained by conventional treatments [13]. There is also an increased likelihood of losing either partial or total activity of specific constitu-

ents of plant extracts when they are isolated or purified even though some of these plant extracts are highly sensitive to the acidic pH of the stomach and may be less effective after ingestion. Because of these obstacles, many of the herbal drugs and extracts are not studied clinically and nanoparticles might be used to address these problems [13, 17]. It should be noted that an adequately high content of the active substance isolated/extracted from the medical plant is vital for producing therapeutic effects in a living organism. Therefore, to overcome poor bioavailability and bioactivity of herbal drug formulations, biotechnological approaches such as nanotechnology and nanomedicine could be a promising innovative technology incorporated into agri-food products for augmenting the level of these phytochemical constituents in plants, to improve the phytotherapy efficiency of herbal drug formulations. The development of a drug delivery system that is authoritative, efficient, and safe for a remedy against illness is a goal for various researchers.

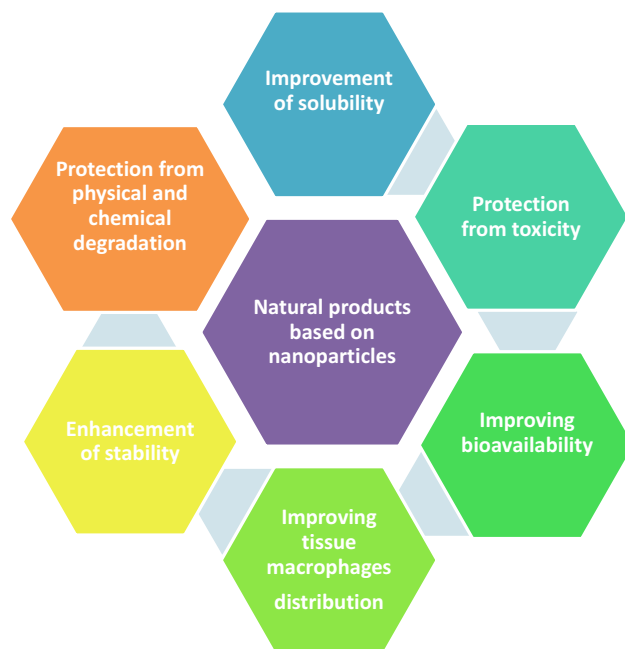


Fig. (1). Application of nanotechnology incorporated into herbal drug formulations for enhancement of efficacy. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

2.1. Herbal Drugs and Nanotechnology

Nanotechnology typically refers to the use of nano-materials between 1 and 100nm in size. These organic nanoparticles are designed in various shapes, sizes, and compositions, and functionalized and modified physiochemically to reach specific properties depending on the features of both the active substance and the targeted organ. Moreover, inorganic nanoparticles, e.g., aluminium oxide, carbon, copper oxide, ceramics, iron oxide, gold, silver, and zinc nanoparticles are also utilized in phytochemical studies [18]. Recent developments in the field of nanotechnology have led to a renewed interest in herbal drug formulations, as a number of nanotechnology approaches have attempted to address phar-

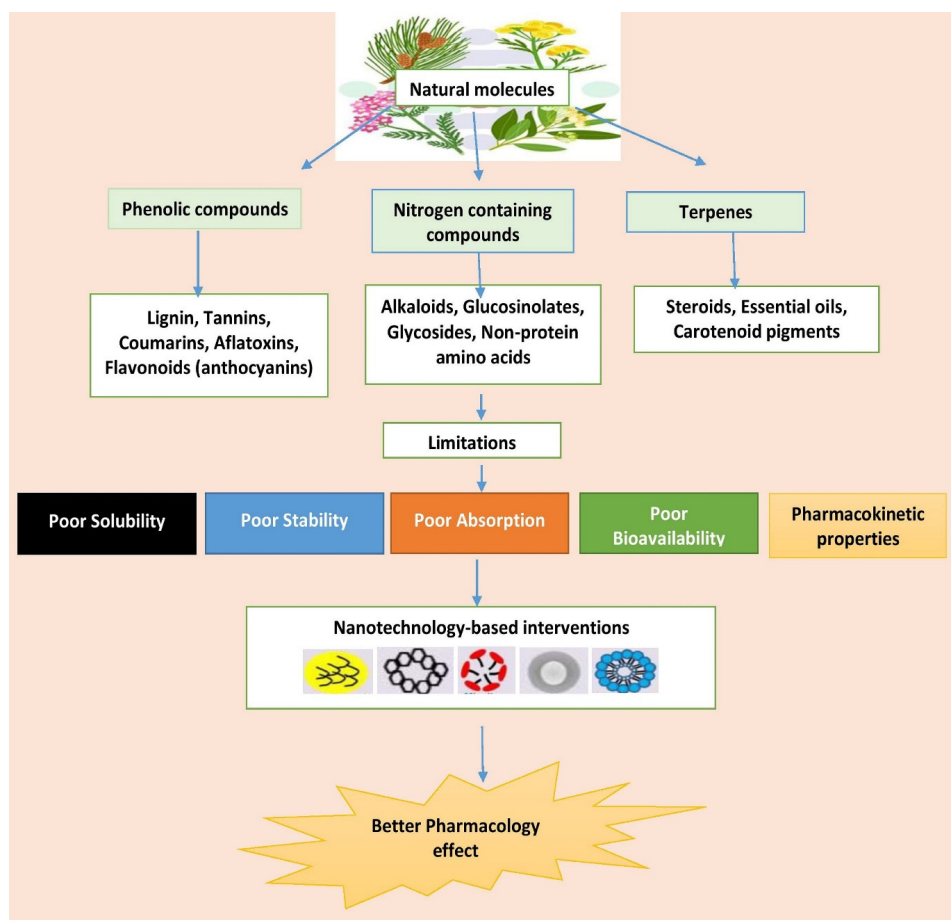


Fig. (2). A schematic indicating the significant impediments related to phytochemicals that limit their application in the clinic. Different methodologies can be utilized to defeat these confinements especially the nanotechnology-based drugs and formulations for targeted therapeutic delivery that may improve the adequacy of natural molecules most likely by expanding their bioavailability and absorption. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

macokinetic properties of different herbal drugs such as liposomes, microemulsions, polymeric nanoparticles, solid lipid nanoparticles (SLNs), liquid crystal systems (LC), and precursors systems for liquid crystals (PSLCs) [13, 3, 19].

Nanoparticle drug delivery systems were initiated to enhance the overall quality of herbal drugs, for instance, zedary turmeric essential oil components have been developed (Fig. 3) to solve several problems associated with the low bioavailability (14). Those are not only enhancing the effectiveness of bioactive compounds by increasing selectivity and efficiency, keeping from thermal- or photo-degradation, decreasing adverse effects but also to reintroduce other agents that were thrown away as they are no longer beneficial [20]. Nanostructures may play a role in formulation action and have the ability to integrate the active ingredient with diverse degrees of lipophilicity and hydrophilicity of drugs as well as delivering to target tissue and organs [21-23].

Besides the advantage of nanotechnology in several aspects of biomedical and pharmaceutical areas, there are also some negative aspects that are associated with using nanotechnologies such as high cost, the difficulty of scaling-up

processes, and inhalation of nanoparticles, which may affect lung diseases, changes in homeostasis, or ultimately even death [3]. Nanocarriers loaded with botanical drug formulation could enhance the action of plant extracts, sustain the release of phytoconstituents, lessen the recommended dose, lower side effects, and boost activity [24, 25]. Table 1 explains the different types of nanomedicine applications in phytopharmaceuticals.

2.2. Herbal Drugs and Liposomes

Liposomes are microscopic vesicles having at least one lipid bilayer and separated by an aqueous medium. Aqueous soluble compounds are encapsulated in the water closet, while adsorbed hydrophobic substances are inserted into the membrane and can be encapsulated too. These carriers composed mainly of phospholipids (synthetic or natural), sterols, and an antioxidant. They are categorized based on their size, the number of lamellae (oligo-, unilamellar or multilamellar, and small (diameters of roughly 25-100 nm), large (diameters of 100 nm to 1 μ m), or giant (diameters greater than 1 μ m) size, and surface charge (anionic, cationic, or neutral) [3]. Liposomes as a novel drug delivery system can enhance the therapeutic activity and safety of many drugs because of

Table 1. Different types of nanomedicine applications in phytopharmaceuticals.

No	Types of Nanomaterial	Plant Sources	Bioactive Compound	Formulations	Functions	Therapeutic Applications	Refs.
1	Emulsion solvent evaporation	<i>Artemisia annua</i> L.	Artemisinin	Artemisinin -loaded nano-particles	Prolonged drug release	Anticancer	[23]
2	Microemulsions	<i>Cinnamomum verum</i> J.Presl	Cinnamon oil	Cinnamon oil formulation with a very low droplet	Drug delivery agents	Wound healing, antibacterial activity	[24]
3	Alginate Nanoparticles	<i>Ocimum sanctum</i> L.	Methanolic extracts	Methanolic extracts of <i>O. sanctum</i> loaded alginate nanoparticles	Biocontrol agent against bacteria in fabrics.	Antibacterial activity	[25]
4	Liposomal	<i>Artemisia arborescens</i> L.	Essential oil	Liposomal incorporation of <i>A. arborescens</i> essential oil	Topical delivery systems	Antiherpetic activity	[26]
5	Liposomes	<i>Cratylia mollis</i> Benth.	Lectin	<i>Cratylia mollis</i> lectin - loaded liposomes	-	Antitumor	[27]
6	Liposomes	egg	Quercetin	lipid thin film formation and extrusion	Drug delivery system	Anxiolytic and cognitive-enhancing effects	[28]
7	Liposomes	<i>Silybum marianum</i>	Silymarin	Silymarin loaded hybrid liposomes	Bioavailability	Stability and <i>in vivo</i> hepatoprotective efficiency	[29]
8	Multivesicular liposome (MVL, namely DepoFoam)	<i>Erigeron breviscapus</i> (Vaniot) Hand.-Mazz.	Flavonoid	Flavonoid isolated from the <i>Erigeron breviscapus</i> (Vaniot) Hand.-Mazz.	Drug delivery system.	Sustained delivery system of breviscapine	[30]
9	Electrospinning	<i>Centella asiatica</i>	Extracts	Gelatin nanofibres containing <i>C. asiatica</i> extracts	Promoting fibroblast proliferation	Skin wounds	[31]
10	Micro-emulsion technique	<i>Curcuma longa</i> L.	Curcuminoids	Encapsulation of curcumin in solid lipid nanoparticles	Sustained delivery of curcumin	Anticancer, antioxidant	[32]
11	Emulsification/evaporation solvent technique	<i>Syzygium cumini</i> L.	Extracts	Polymeric NP formulation of <i>Syzygiumcumini</i>	Drug delivery systems.	Antidiabetic	[33]
12	Solid lipid nanoparticles	<i>Salvia officinalis</i> L., <i>Satureja hortensis</i> L.	Phenolic compounds	Witepsol and carnauba loaded extracts	Vehicles	Antioxidant activity	[34]
13	Dialysis method	<i>Camptotheca acuminata</i> Decne.	Camptothecin	CPT encapsulated nanoparticles	Drug delivery system.	Anticancer	[35]
14	Nano-carrier transdermal gel	<i>Curcuma longa</i> L.	Curcumin	Diclofenacdiethylamine and curcumin nanocarrier transdermal gel	Drug delivery system.	Inflammation	[36]
15	Nanoprecipitation	Plants and foods	Oleanolic acid	Oleanolic acid loaded PEGylated PLA and PLGA Nanoparticles	Drug delivery system.	Anticancer	[37]

(Table 1) contd...

No	Types of Nanomaterial	Plant Sources	Bioactive Compound	Formulations	Functions	Therapeutic Applications	Refs.
16	Nanoprecipitation technique	<i>Cuscuta chinensis</i> Lam.	Flavonoids and lignans	Nanoparticles of <i>Cuscuta chinensis</i>	Drug delivery agents.	Hepatoprotective and antioxidant effect	[38]
17	Rotary-evaporated film ultra-sonication method	<i>Glycyrrhiza glabra</i> L.	Glycyrrhizic acid and glabridin	Glycyrrhizic acid loaded nanoparticles	Drug delivery system.	Anti-inflammatory, antihypertensive	[39]
18	Marinosomes	<i>Curcuma longa</i> L.	Curcumin	Encapsulated curcumin in krill lipids-based liposomes	Sustained release	Antioxidant and anticancer activity	[40]
19	Nanostructured lipid carriers	<i>Thymus vulgaris</i> L.	Thymol	Thymol in nanostructured lipid carrier's formulation	Topical delivery	Inflammatory skin diseases (wound)	[41]
20	Ethosomes	<i>Tripterygium wilfordii</i> Hook. f.	Triptolide	Triptolide-loaded ethosomes	Topical application	Improved the percutaneous permeation	[42]
21	Ethosomes	<i>Sophora flavescens</i> Aiton.	Matrine	Matrine ethosome	Topical application	Improved the percutaneous permeation, anti-inflammatory activity	[43]
22	Mesoporous silica nanoparticles	<i>Curcuma longa</i> L.	Curcumin	Curcumin-mesoporous silica nanoparticles	Bioavailability	Anti-inflammation	[44]
23	Liposome	<i>Scutellaria baicalensis</i> Georgi	Baicalin	Baicalin -loaded liposome	Drug delivery system	Bioavailability	[45]
24	Elastic liposomes	<i>Camellia sinensis</i> (L.) Kuntze	(+)-catechin	(+)-catechin loaded in liposomes	Drug delivery system	Oral bioavailability	[46]
25	Emulsification-solvent evaporation	<i>Curcuma longa</i> L.	Curcumin	Curcumin-loaded PLGA nanoparticles	Penetration	Brain disease	[47]
26	Nano-precipitation	<i>Curcuma longa</i> L.	Curcumin	Encapsulated curcumin in nanoparticles	Drug delivery system	Anti-cancer	[48]
27	Liposomes	<i>Curcuma longa</i> L.	Curcumin	Curcumin-containing liposomes	Drug delivery system	Type 2 diabetes	[49]
28	Solid lipid nanoparticles	<i>Curcuma longa</i> L.	Curcumin	Curcumin-solid lipid nanoparticles	Bioavailability	Asthma therapy	[50]
29	Polymer-based nanoparticle	<i>Curcuma longa</i> L.	Curcumin	Biodegradable and biocompatible polymer.	Drug delivery system	Alzheimer's disease	[51]
30	Nanoprecipitation	Grapes, peanuts, red wine	Resveratrol	Resveratrol-loaded nanoparticles	Sustained release	Malignant glioma therapy	[52]
31	Liposome	<i>Salviae miltiorrhizae</i> (Danshen)	Salvianic acid	Liposome Formulation of Salvianic Acid	Efficacy	Bone fracture union	[53]
32	Poly (caprolactone)-poly (ethylene glycol)- Poly (caprolactone) (PCL-PEG-PCL)	Broccoli, cabbage and kale	Sulforaphane	Encapsulated sulforaphane by PCL-PEG-PCL nanoparticles Or PCL-PEG-PCL micelles	Drug delivery system	Anti-cancer	[54]

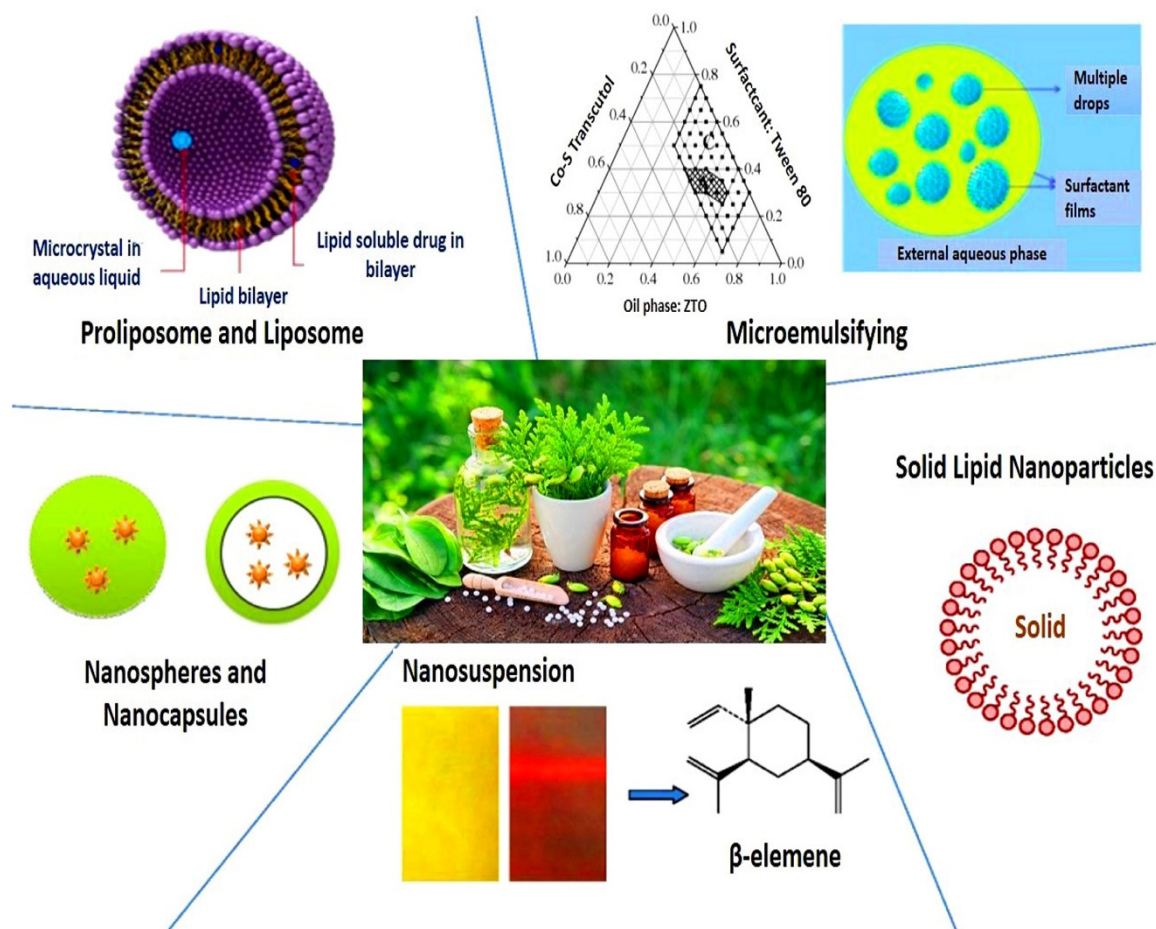


Fig. (3). Nanoparticle drug delivery systems for herbal drug formulations to enhance their bioavailability and absorption modified from (14). (A higher resolution / colour version of this figure is available in the electronic copy of the article).

their ability to deliver the entrapped drugs across cell membranes and to protect their content from ecological exposure. They play a vital role in antiviral agents as they can be administered parenterally for diffused systemic infections, in drops for herpes keratitis, and topically applied on lips and mucous membranes in the form of gel or cream and this depends on the type and severity of the infection [26].

Liposomal formulations with extracted essential oil from *Artemisia arborescens* L. (Asteraceae), *in vitro* showed enhanced antiherpetic activity against one of the most common viral diseases in humans, Herpes simplex virus (HSV-1), by penetrating the barrier of the cytoplasmic viral in the antiviral assay. This clearly demonstrates that liposomes strategy can benefit in promoting the targeting of essential oils to cells and in transferring active substances against viral diseases that cannot pass or have difficulty crossing the cytoplasmic barrier [26]. The antitumor activity of *Cratylia mollis* Mart-loaded liposomes, obtained by the lipid thin film method, was investigated *in vivo* against Sarcoma-180 in Swiss mice [27] which showed an increase in activity by 28% compared to free Cra solution as well as a decrease in its toxic effect on the hepatoprotective. Quercetin is a flavanol compound derived from the flavonoid group of polyphenols. Quercetin was loaded with intranasal liposomes to test

anxiolytic and cognitive activities utilizing the elevated plus maze and the Morris water maze in male Wistar rats. The method used was lipid thin film formation (the mixture of egg phosphatidylcholine (EPC) and cholesterol (Chol)), and extrusion, with 200 nm mean diameter and negative surface charge. The results demonstrated anxiolytic and cognitive-enhancing effects probably *via* alteration of neurotransmitters such as gamma-aminobutyric acid (GABA) and serotonin; thus intranasal quercetin liposomes could deliver quercetin to the central nervous system [28].

Silymarin is a mixture of flavonolignans from seeds of the milk thistle *Silybum marianum* (Compositae) and has been known for various therapeutic applications, but due to low oral bioavailability, it is weakly absorbed (20%–50%) from the digestive system [29]. Silymarin-encapsulated hybrid liposomes consisting of lecithin (L), cholesterol (Ch), stearyl amine (SA), and Tween 20 (T20) in a molar ratio of (9:1:1:0.5) were prepared for buccal administration for its hepatoprotective activity. This technique displayed a substantial reduction in both transaminase levels (enzymes that catalyze the transfer of an amino group from an amine to an acceptor, important in the production of various amino acids which form proteins) when challenged with CCl₄ (intraperitoneally) compared to orally taken silymarin sus-

pension in albino rats. Breviscapine, a well-known bioactive substance from *Erigeron breviscapus* (Compositae) claimed to treat ischemic cerebrovascular and cardiovascular diseases, has been studied to extend the interval of the drug in the blood circulation, decrease the frequency of injection administration, and thereafter, afford patient compliance. A multivesicular liposome (MVL, namely DepoFoam) was synthesized as a sustained drug delivery system for breviscapine. *In vitro* and *in vivo* pharmacokinetics of MVLs containing breviscapine (bre-MVLs) showed a positive result for prolongation of the drug [30].

2.3. Herbal Drugs and Microemulsions

Microemulsions (MEs) are low-stickiness, visually isotropic, and thermodynamically steady colloid systems; they are diaphanous emulsions, in which oil material is disseminated in a water medium (or vice-versa) including a surfactant, with or without an appropriate cosurfactant. MEs probably carry the natural molecules when they are solubilized in the aqueous or oil phases. They have been employed extensively in a variety of pharmaceutical products as drug delivery carriers for oral, parental, transdermal, rhinal, intravenous, and dermal drug delivery modalities [55, 56].

MEs can enhance solubility with differing degrees (hydrophilicity/lipophilicity) and stability of many drugs, being able to convey active molecules within the same formulation [21]. Curcuminoid was isolated from *Curcuma longa* L. and incorporated into microemulsion for studying the inhibition mechanism of HT-29 colon cancer cells. The preparation was made of soybean oil, Tween 80, ethanol, and water with high stability and particle size of (10.9 nm), zeta-potential of -65.2 mV, and encapsulation efficiency of 85.7%. Curcuminoid extract plus microemulsion could inhibit cancer cell growth with the IC_{50} being 3.83 and 2.51 $\mu\text{g mL}^{-1}$ after 24 h of incubation in an effective manner, respectively, while by increasing incubation time to 48 h, the inhibition was reduced to 2.23 and 1.94 $\mu\text{g mL}^{-1}$ [57].

β -Caryophyllene is a bicyclic sesquiterpene essential oil of *Copaifera multijuga* (copaiba), which has been used widely in Brazilian traditional medicine. Two microemulsion methods were prepared (% w/w), firstly-I containing (47.1 water, 8.5 Plurol Oleique, 33.8 Labrasol and 10.6 copaiba oil-resin) and secondly-II containing (39.0 water, 18.3 Plurol Oleique, 36.6 Labrasol and 6.1 copaiba oil-resin). The second one showed a much stronger antibacterial activity [58].

2.4. Herbal Drugs and Polymeric Nanoparticles (PNPs)

Polymeric nanoparticles (PNPs) ranging from 10–1,000 nm in diameter are nano-spheres and nanocapsules in the form of a solid phase that is used to control drug release, protect drug molecules, and target the specific tissues. Polymeric nanoparticles also work as a cell surface, and therapy and imaging enhancements in the therapeutic index. [59, 3]. The fundamental features of the polymeric nanoparticle system are biodegradability, non-toxicity, biocompatibility, protracted circulation, and a wide range of curative substances. They can also improve the solubility and absorption of phytomolecules and reduce the therapeutic dose compared to

conventional formulations [60]. They can be synthesized from natural or artificial, biodegradable polymers with the preference of natural materials due to having more beneficial aspects. For instance, they can transfer more than one biologically active substance with the same vehicle, augmenting the remaining period in the target tissue, thus offering a continued release system, and diminishing adverse effects [61]. Nanoparticles are useful in the blood system, as they are steady, nonthrombogenic, nonimmunogenic, non-toxic, non-inflammatory, do not activate neutrophils, and avoid the reticuloendothelial system [13, 61]. Nanoscale systems are also known as submicrometer with the diameters smaller than 1 μm , their curative applications are the way of taken *via* the mouth, position privacy, and increased pharmaceutical effect. In ophthalmic administration, nanoparticles could increase ocular bioavailability by controlling the release of active compounds and reducing side effects [3]. Polymeric nanoparticles can appear in the form of nanocapsules (NCs) and nanospheres (NSs), with various structures and compositions. Nanocapsules include an oily core surrounded by a polymeric membrane; the phytoconstituents can be adsorbed to the polymeric membrane or dissolved in the oily core. While nanospheres are matrix systems where the drug is retained or adsorbed and physically and uniformly dispersed [62, 59]. The popular methods to produce polymeric nanoparticles are widely categorized as follows: poly (lactic acid) (PLA), poly (cyanoacrylate) (PCA), poly (D, L-lactide-co-glycolide) (PLGA), and poly (D, L-glycolide) (PLG) involving these methods nanoprecipitation, solvent evaporation, Emulsion (mini and micro), emulsification/solvent diffusion, interfacial polymerization, salting out, controlled/living radical polymerization (C/LRP), dialysis, and supercritical fluid technology (SCF) [59].

Nowadays, considerable progress has been made concerning involving medicinal plants in nanotechnological processes. It has been reported [25] that methanol extract from *Ocimum sanctum* L. (Lamiaceae) loaded into sodium alginate chitosan nanoparticles (OSN) exhibited better antimicrobial activity against *Escherichia coli*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, and *Bacillus subtilis* with excellent wash durability other than unloaded formulation. Curcumin-loaded polymeric nanoparticles were made based on a wet-milling technique with a narrow particle size distribution in the range of 2–40 nm, without changing the chemical structure of nanocurcumin during encapsulation, which was identical to the original curcumin molecule. The formulation showed a much better solubility in water in the absence of any surfactants, as well as an improved antibacterial activity using a microplate dilution technique against several microorganisms such as *S. aureus*, *B. subtilis*, *E. coli*, *P. aeruginosa*, *Penicillium notatum*, and *Aspergillus niger*. Then the mechanism of bacterial inhibition of curcumin nanoparticles was studied by Transmission Electron Micrograph (TEM) analysis, and it was observed that these nanoparticles could easily enter the bacterial cell, breaking the cell wall and inducing apoptosis [63].

One of the major challenges in chemotherapy is delivering the active substance to target the affected area in a dose so that it leads to optimum therapy. For this, drug-formulated polymeric nanoparticles have been found useful in preclinical

cal and clinical studies against tumor cells due to their ability to minimize toxicity and their protective agent to block the interaction between the active constituents and healthy cells [63, 64]. Formulated PLGA encapsulated nanoparticles of root extract of *Phytolacca decandra* (Phytolaccaceae) (NPD) which were stabilized by F68 (polyoxyethylene-polyoxypropylene) improved drug bioavailability, and thereby showed a better chemo-preventive action against lung cancer *in vivo* and on A549 cells *in vitro* than that of PD [65]. Curcumin is a yellow color polyphenol substance isolated from rhizomes of *Curcuma longa* (Zingiberaceae); it has exerted potent molecular targets for cancer control in the investigation involving human tumor cells and animal models of carcinogenesis. It is widely used as an ingredient in the herbal supplement, cosmetics, food flavoring, and food coloring. It is non-toxic and highly potent and has been used to heal several ailments as an alternative drug, but due to lower aqueous solubility and bioavailability when administered orally, its clinical applications are limited. Polymeric nanoparticles have been partially addressed, especially for hydrophobic bioactive constituents such as curcumin.

Researchers synthesized nanocurcumin *via* loaded polymeric nanoparticles, using aggregated structures containing randomly crosslinked copolymers of N-isopropylacrylamide, N-vinyl-2-pyrrolidone, and poly (ethylene glycol) monoacrylate. Nanocurcumin has been used for antitumor activity *in vitro*, using different human pancreatic carcinoma cell lines, assured by cell viability and clonogenic assays. Nanocurcumin has potential use in clinical trials, facilitating easy dispersal in aqueous media. The possible mechanism of action of nanocurcumin is due to releasing curcumin in free form, inducing apoptosis, blocking the activation of nuclear factor kappa B (NFkB), and regulating levels of proinflammatory cytokines, such as interleukin 6, interleukin 8, and the tumor necrosis factor [66]. Encapsulation of curcumin in poly (lactic-co-glycolic acid) (PLGA) nanospheres, with the efficiency of $90.88 \pm 0.14\%$, the size range of 35 nm to 100 nm, and mean particle size of 45 nm, using solid/oil/water emulsion solvent evaporation method, was studied against prostate cancer cell lines. Curcumin-loaded PLGA nanospheres showed proper dispersion in aqueous media with a remarkable effect on the cancer cells as compared to free curcumin [32].

2.5. Herbal Drugs, Solid Lipid Nanoparticles and Nanostructured Lipid Carriers

Solid Lipid Nanoparticles (SLNs) are colloidal carrier systems that integrate other colloidal systems for drug delivery, while averting, or diminishing, some of their disadvantages [67]. SLNs are prepared by various techniques including ultrasonication, emulsification-solvent evaporation, emulsification-solvent diffusion, phase inversion, microemulsion, and solvent displacement [68]. They are known for higher physicochemical constancy, with better protection against degradation of labile drugs, which can be readily synthesized on a large scale [67, 69]. These particles are comprised of highly purified triglycerides (mainly of lipids) that are solid at room temperature and produced from solid lipids, or mixtures thereof, and stabilized by surfactants [70].

The formation of SLNs can be promoted by adding a liquid lipid to an oil/water emulsion containing either a solid lipid or a mixture of them [69]. Once the system is ready, crystallization occurs which leads to low encapsulation efficiency and drug release [71]. SLNs may be administered in the pharmaceutical industry in different routes, such as oral, parenteral, and percutaneous, because of their small size (50-1,000 nm) and biocompatibility [72].

Nanostructured Lipid Carriers (NLCs) are second-generation lipid nanoparticles consisting of a mixture of spatially diverse lipid molecules, *i.e.*, a solid lipid mixed with a liquid lipid [73]. NLCs ameliorate the efficacy of encapsulation and minimize the expulsion of active particles during encapsulation [3]. These formulations may be administered *via* the oral, pulmonary, intravenous, and dermal route [71]. NLCs enhance the oral bioavailability of various encapsulated drugs *via* selective uptake through the lymphatic route or the patch's patches [73]. Both SLNs and NLCs can be produced by several techniques such as emulsification-sonication, high-pressure homogenization, microemulsion and solvent emulsification-evaporation method [74]. Quercetin is a natural polyphenol from the flavonoid group with various bioactivities and pharmacological actions. Quercetin has low hydrophilicity and absorption with weak oral bioavailability. Thus, it is extremely important that to what extent it can be absorbed after ingestion, considering distribution in the human body tissue. Quercetin-loaded solid lipid nanoparticles (QT-SLNs) were prepared by emulsification and low-temperature solidification method as an oral delivery carrier for poorly water-soluble quercetin drugs with an average diameter of 155.3 nm. QU, glyceryl monostearate, and soy lecithin were mixed with the solvent (chloroform and acetone in 1:1 ratio v/v). QU-SLN after oral administration exhibited enhanced bioavailability relatively five-fold in the intestine (ileum and colon), rather than the stomach, compared to free QU *in vivo* [75]. Additionally, *in vitro* and *in vivo* skin permeation study of quercetin-loaded nanostructured lipid carriers (QT-NLCs) as a potential topical delivery system was conducted using the previous method with the average particle size of 215.2 nm. QT-NLCs could promote the permeation of QT, increase the amount of QT retention in the epidermis and dermis compared to QT propylene glycol solution, and boost the effect of anti-oxidation and anti-inflammation exerted by quercetin [76].

For optimum topical delivery of the flavonoid quercetin, Nanostructured Lipid Carrier (NLC) systems were developed using the probe ultrasonication method, where a part of the solid lipid (glyceryl dibehenate) in the SLN formulation was substituted with a liquid lipid (oleic acid). The mean particle size was around 282 nm, with adequate physical stability for up to 14 weeks at 2-8 °C. *In vitro*, the NLC system displayed the highest enhancement in the topical delivery of quercetin, which retained in full-thickness human skin, compared to a control with the same features [77]. Curcumin was loaded into solid lipid nanoparticles (C-SLNs) with a particle size of 134.6 nm using microemulsification. *In vivo* pharmacokinetic study was carried out after an oral administration of C-SLNs (50, 25, 12.5 and 1 mg/kg dose). A significant improvement was observed in oral bioavailability 155 times at 1 mg/kg compared to free Curcumin [78].

2.6. Herbal Drugs and Liquid Crystalline Nanoparticles (LIPs)

Liquid crystals are a featured stage of condensed materials that lie between an atomic ordered crystalline solid and an isotropic liquid in an intermediate state. They may be ordered or disordered, as evidenced by their ease of efflux. LCs are classified according to, firstly, Thermotropic Liquid Crystals (TLC) at which the liquid crystal becomes an isotropic liquid with a molecule that forms the mesophase. This mesophase formation is dependent on concentration, solvent, and temperature and, secondly, Lyotropic Liquid Crystals (LLC). LLC is composed of amphiphilic molecules including small polar (hydrophilic) and a large non-polar (hydrophobic) tail [3, 79, 80]. These distinct features provide the possibility of dissolving and dispersing of various natural compounds based on their polarity (water-soluble and insoluble) [81]. Liquid crystalline nanoparticles (LIPs) have been renowned, as a potential oral drug delivery vehicle, imparting oral absorption, maintenance of chemical stability, sustained release profile, protection to the loaded drug, skin penetration enhancement, *etc.* [82]. LCs effectively promote interaction between the drug and cell membranes that are normally inactive [81].

Plant oils or vegetable oils have been found useful for developing LC systems, due to their favorable properties, including low viscosity and low molecular weight, providing optimum for skin penetration, and allowing a growing load of the curative agent [83]. Plant oils have been exploited increasingly in all types of cosmetic products due to consumer demand, as they are known to be less occlusive than petroleum-derived emollients [84]. One issue that should be considered is that their chemical composition is inconsistent and may cause stability problems as well [84]. One of the most studied types of plant oils is essential oils which are known to have many biological activities [85]. However, natural products have been used to form LC systems, but there are few reports concerning incorporating LC to herbal formulations as drug delivery.

CONCLUSION

Phytomedicine application plays a vital role in public healthcare worldwide. The treatment of any illness could be ameliorated either by producing new drugs or the development of current drugs with more effective and safer use. Herbal drugs and natural products are getting popular throughout the world, but their limits in clinical applications are due to poor solubility, poor oral bioavailability, low permeation, instability, and pharmacological property. Consequently, the merger of herbal drugs with nanocarriers could be an alternative strategy to improve the efficiency of curing various ailments.

Nanotechnology may be useful as a drug vehicle for herbal medicine, to enhance the low bioavailability and poor bioactivity of active natural molecules present in plant materials, even though their exploitation is still limited especially in clinically viable therapy trials. There is still a lack of toxicological profiling and inadequate data on the safety and toxicity of nanomaterials and nanomedicines is available, thus potential hazards associated with nanotechnology-based

drug delivery systems should also be considered while incorporating them into the botanical drug formulation. The nanomedicine approaches could greatly potentiate the low bioavailability and poor pharmacokinetic profile of natural compounds and ameliorate the pharmacodynamic properties of different drugs.

AUTHOR CONTRIBUTION'S STATEMENT

HA has designed and written the first draft of the manuscript. All authors have read and revised critically, and approved the final submitted version of the manuscript.

CONSENT FOR PUBLICATION

All authors agreed to the final version of the manuscript. The work is the authors' original work and has not been previously published.

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CONFLICT OF INTEREST

The authors declare that the research was carried out in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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